

ESPRESSIF · WI-FI 6 · RISC-V

ESP32-C6

Wi-Fi Capabilities

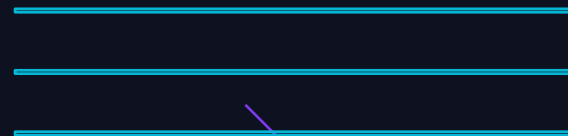
*Wi-Fi 6 (802.11ax) · Station & Access Point
Security · Provisioning · Real-World Use*

Wi-Fi 6

802.11ax

WPA3

TWT



6

ESP32-C6 at a Glance

The first ESP32 with Wi-Fi 6 (802.11ax) — delivering higher throughput, lower latency, and better crowded-network performance.

Wi-Fi 6

802.11ax

2.4 GHz

frequency band

~150 Mbps

peak throughput

WPA3

latest security



OFDMA

Splits channels to serve multiple clients simultaneously — 3× better efficiency in dense networks



WPA3 + OWE

Simultaneous Authentication of Equals (SAE) + Opportunistic Wireless Encryption for open networks



Target Wake Time (TWT)

Devices negotiate precise sleep/wake schedules with the AP, cutting idle power by 10×+



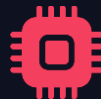
BSS Coloring

Spatial reuse tag reduces inter-network interference in dense environments like offices and apartments



STA + AP Concurrent

Run Station and Access Point simultaneously — bridge to cloud while serving local clients



RISC-V Core

32-bit single-core RISC-V at 160 MHz — deterministic execution, open ISA, low power

Wi-Fi 6 vs Previous Generations

Wi-Fi 6 (802.11ax) delivers meaningful gains across every dimension relevant to IoT deployments.

Feature	Wi-Fi 4 (n)	Wi-Fi 5 (ac)	Wi-Fi 6 (ax) C6
Standard	802.11n	802.11ac	802.11ax
Max rate (2.4G)	~150 Mbps	N/A (5 GHz only)	~150 Mbps + MU
Multi-user DL	None	MU-MIMO (DL)	OFDMA + MU-MIMO
Multi-user UL	None	None	OFDMA UL
Power saving	DTIM	DTIM	TWT (precise)
Security	WPA2	WPA2	WPA3 + OWE
Spatial reuse	None	None	BSS Coloring
IoT suitability	Good	OK (5 GHz gap)	Excellent

ESP32-C6 is the only ESP32 with Wi-Fi 6 — a key differentiator for dense IoT and smart-home deployments.

Wi-Fi Operating Modes

Three modes — use them individually or all at once.



Station (STA)

Connect to a router or AP

- Gets IP from router DHCP
- Supports static IP config
- WiFiMulti for multiple SSIDs
- Auto-reconnect on drop

```
WiFi.begin("MySSID", "pass");
while (WiFi.status() != WL_CONNECTED)
  delay(500);
Serial.println(WiFi.localIP());
```



Access Point (AP)

Host a hotspot; clients connect

- Custom SSID + password
- Up to 10 client connections
- Built-in DHCP server
- Hidden SSID option

```
WiFi.softAP("ESP32C6_AP", "secret");
IPAddress ip = WiFi.softAPIP();
Serial.println(ip);
```



STA + AP (Dual)

Both modes simultaneously

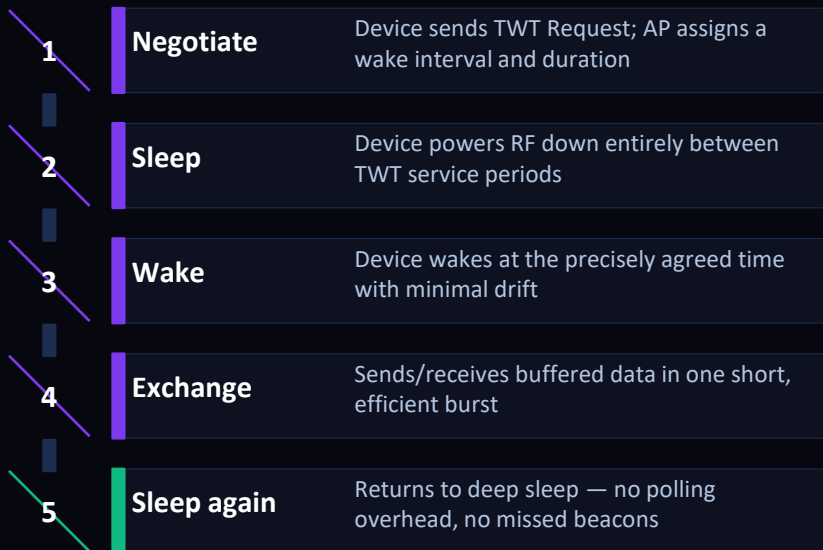
- Cloud uplink + local config
- Provisioning portal pattern
- Separate IP per interface
- Wi-Fi bridge / extender

```
WiFi.mode(WIFI_AP_STA);
WiFi.begin("UpstreamSSID", "pw");
WiFi.softAP("LocalAP", "pass");
```

Target Wake Time (TWT)

TWT lets each device negotiate a private sleep/wake schedule with the AP — slashing idle current by 10x+ versus DTIM polling.

How TWT Works



Power Comparison

Always-on Active

80 mA

DTIM1 (Wi-Fi 4/5)

12 mA

TWT (Wi-Fi 6)

1.2 mA

Enable TWT (esp-idf)

```
esp_wifi_set_config(WIFI_IF_STA,  
    &(wifi_config_t){.sta.twt = {.setup_cmd=TWT_REQUEST}});
```

Wi-Fi Security Protocols

ESP32-C6 supports WPA3 — the strongest Wi-Fi security standard — alongside full mbedTLS for HTTPS/TLS.



Open

No auth

Avoid

No encryption — all traffic visible



WEP

1999

Broken

RC4 cipher cracked in seconds



WPA

2003

Legacy

TKIP — improved but still weak



WPA2

2004

Current

AES-CCMP — strong; widely deployed



WPA3

2018

Recommended

SAE handshake + forward secrecy

WPA3 on ESP32-C6

SAE prevents offline dictionary attacks | Perfect Forward Secrecy | OWE for open nets

HTTPS / TLS 1.2 & 1.3

mbedTLS built-in — encrypt all cloud and API communications end-to-end

Wi-Fi Provisioning Methods

Getting Wi-Fi credentials onto devices securely — without hardcoding them.



SmartConfig

Phone app broadcasts credentials over multicast UDP. ESP32-C6 snoops encrypted packets to extract SSID/password.

- ✓ No extra hardware
- ✓ Espressif app support
- ✓ Fast (under 10 s)

```
WiFi.beginSmartConfig();
while (!WiFi.smartConfigDone())
  delay(500);
WiFi.stopSmartConfig();
```



SoftAP Portal

C6 starts a temporary hotspot; user connects and enters credentials via captive web page. Most universal method.

- ✓ Works on any device
- ✓ Visual feedback UI
- ✓ WiFiManager library

```
#include <WiFiManager.h>
WiFiManager wm;
wm.autoConnect("C6_Setup");
```



BLE Provisioning

Credentials delivered over Bluetooth LE using Espressif unified provisioning API. Seamless for mobile apps.

- ✓ No AP mode needed
- ✓ ESP Rainmaker ready
- ✓ Secure BLE channel

```
WiFiProv.beginProvision(
  WIFI_PROV_SCHEME_BLE,
  WIFI_PROV_SECURITY_1,
  "abcd1234");
```

TCP/IP Stack & Protocols

Full lwIP TCP/IP stack in ESP-IDF — ready for HTTP, MQTT, WebSocket, mDNS and more out of the box.

ESP-IDF lwIP Stack

Application Layer

HTTP · MQTT · WebSocket · mDNS

Transport Layer

TCP · UDP · TLS 1.2 / 1.3

Network Layer

IPv4 · IPv6 · ICMP · DHCP · DNS

Link / MAC Layer

802.11ax frame control & queuing

PHY (2.4 GHz RF)

OFDMA · Wi-Fi 6 · DSSS / OFDM

Key Application Protocols

HTTP

HTTP / HTTPS

REST APIs, OTA, web config portals



MQTT

Lightweight pub/sub for cloud IoT



WebSocket

Full-duplex real-time data streams



mDNS / DNS-SD

Zero-conf device discovery (Bonjour)

Real-World Applications

Wi-Fi 6 unlocks scenarios where previous ESPs struggled — dense deployments, long battery life, secure enterprise networks.

Embedded Web Server



Host REST APIs, dashboards and OTA endpoints directly from the chip at Wi-Fi 6 speeds

Cloud IoT (MQTT/HTTPS)



Push telemetry to AWS IoT, Google Cloud or Azure over TLS with WPA3 security

Smart Home / Matter



Works with Home Assistant and Matter; Wi-Fi 6 handles many devices on crowded home networks

Battery Sensors (TWT)



TWT lets coin-cell sensors run for months — no frequent charging, no compromises on reporting

Dense IoT Deployments



OFDMA serves dozens of devices simultaneously — perfect for offices, factories, smart buildings

OTA Firmware Updates



Roll out firmware wirelessly via ArduinoOTA or esp-idf OTA partitions with rollback support

Summary

Standard	Standard	802.11ax (Wi-Fi 6) · 2.4 GHz · ~150 Mbps
Wi-Fi 6 Gains	Wi-Fi 6 Gains	OFDMA multi-user · BSS Coloring · 4x efficiency vs Wi-Fi 4
TWT	TWT	Target Wake Time — 10x+ power reduction for sleepy IoT nodes
Modes	Modes	Station · Access Point · STA+AP concurrent
Security	Security	WPA3 (SAE + OWE) · WPA2 · mbedTLS for HTTPS/TLS
Provisioning	Provisioning	SmartConfig · SoftAP portal · BLE provisioning (unified API)

Docs: docs.espressif.com/esp32-c6 · github.com/espressif/esp-idf · espressif.com